PROJECT PROPOSAL

TECHNOLOGY CAPSTONE PROJECT

MIS 4500

PROJECT TITLE:

Personal Health Record Management and Sharing System

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CHAPTER 01

INTRODUCTION

Handling personal health records is essential to ensure they are properly kept and used for medical advice and information. Patients may receive different medical reports, prescriptions, and other health documents from different health care providers, making it difficult to keep these documents organized and available when needed. This project aims to design a PHRMS (Personal Health Record Management and Sharing System) using web technology, which would enable patients to manage their health information, submit documents for administrative review, and share authorized health information with their physicians. The system will be developed as a functional academic prototype using fictitious data.

AIM AND OBJECTIVES OF THE PROJECT

AIM: To design, develop, and evaluate a web-based Personal Health Record Management and Sharing System that will facilitate patients to organize their medical information, to control document reviews, and to share authorized documents securely with doctors.

OBJECTIVES

1. To review the patient, doctor, and assigned document review team's needs.
2. To create a user-friendly web application having different user interfaces and role-based permissions.
3. To use medical document submission, administrative review, and approval workflows.
4. To create a chronological patient health record and Doctor access controlled by the patient.
5. To develop a summary that can be used as a consultation document with selected patient records.
6. Test the system to determine its functionality, access, and ease of use with make-believe data

PROBLEM STATEMENT

Patients may have to keep records of medical documents, such as medical reports, prescriptions, consultation notes, etc., from various health care providers. It can be difficult to keep these records well organized and have important information accessible when consulting with a patient. Furthermore, the system needs to provide access to health information to a number of stakeholders, with clear permissions and a clear procedure for handling the documents supplied.

The aim of this project is to overcome these challenges by creating a web application that incorporates document submission, administrative review, chronological health record management, and patient-authorized information sharing. However, the prototype will showcase these features in a non-commercial setting, not to mention a clinical setting or the integration of real healthcare institutions.

COMPARATIVE ANALYSIS OF SIMILAR SYSTEMS / SOLUTIONS

Current digital health platforms have shown that personal health information management and information sharing are well-known and proven capabilities. HealthSync refers to the document upload and sharing of health records, while NexusCare and Medica refer to digital health record services and/or healthcare functions.

Table 1.1: Comparative Analysis of Existing Systems

Feature	**HealthSync**	**NexusCare**	**Medica**	**Proposed System**
Health record management	Available	Available	Available	Planned
Medical document submission	Available	Not confirmed	Not confirmed	Planned
Record sharing and access management	Available	Role-based access described	Not confirmed	Patient-authorised access
Administrative document review workflow	Not confirmed	Not confirmed	Not confirmed	Planned
Chronological health history	Not confirmed	Not confirmed	Visit history described	Planned

Consultation-ready summary	Not confirmed	Not confirmed	Not confirmed	Planned

Source: *Official platform descriptions and HealthSync help pages. "Not confirmed" means the reviewed sources did not establish the feature; it does not mean the platform lacks it.*

The comparison indicates that the proposed project is not based on the claim that personal health records or sharing functions are new. Its contribution is the implementation and evaluation of an integrated workflow connecting document submission, administrative review, record organisation and authorised doctor access.

PROPOSED SOLUTION

The solution proposed is a working Web application that will have three different stakeholder roles: patient, doctor, and designated document review team. Every role will have its own interface and permissions based on the role's responsibilities. The patient can provide medical files, add health data, check the status of their records, and control who may access their medical records. The review team will review submissions, approve, reject, and/or seek corrections prior to records being added to the patient's approved health history. Physicians will have access to records as agreed by the patient and be able to schedule consultations with the appropriate information. The system will also include a chronological record view and a consultation-ready summary based on selected records. The review will be of an administrative nature and will not be used to confirm the clinical authenticity of a document.

Figure 1.1: Proposed Document Review and Health Information Sharing Workflow



Source: *AI-generated illustration created using OpenAI ChatGPT (2026).*

SCOPE

IN SCOPE

- Patient, doctor, and document review team interfaces.
- User authentication and role-based permissions.
- Medical document upload and manual health information entry.
- Administrative document review, approval, rejection and correction requests.
- Chronological display of approved health records.
- Patient-authorised doctor access and basic access logging.
- Consultation-ready summaries based on selected records.
- Functional and usability testing using fictitious patient data.

OUT OF SCOPE

- Integration with real hospitals, laboratories, or national health databases.

- Clinical diagnosis, treatment recommendations or automated medical decisions.
- Independent verification of medical document authenticity.
- Use of real patient information in the academic prototype.
- Claims of production-level security, regulatory certification, or deployment as a live healthcare service.

FEATURES AND FUNCTIONALITIES

FUNCTIONAL REQUIREMENTS

ID	Requirement
FR01	The system shall allow users to log in according to their assigned roles.
FR02	Patients shall be able to upload medical documents and enter health information.
FR03	The review team shall be able to approve, reject or request corrections for submitted documents.
FR04	The system shall display submission status to the patient.
FR05	Approved records shall be organized chronologically in the patient profile.
FR06	Patients shall be able to authorize or revoke doctor access to their records.
FR07	Doctors shall be able to view records they are authorized to access.
FR08	The system shall generate a consultation-ready summary from selected records.
FR09	The system shall record relevant access and review actions.

NON-FUNCTIONAL REQUIREMENTS

ID	Requirement
NFR01: Usability	The interface shall be clear, consistent, and easy to navigate.
NFR02: Security	The system shall enforce authentication and role-based access restrictions.
NFR03: Privacy	Only authorized users shall be able to access permitted health information.
NFR04: Reliability	The system shall handle valid submissions and review actions consistently.
NFR05: Performance	Common operations shall respond within an acceptable time under normal prototype usage.
NFR06: Maintainability	The application shall use a structured codebase that can be tested and modified.

NFR07: Data Integrity	The system shall maintain consistent record statuses and prevent unauthorized modifications.

DELIVERABLES OF THE PROJECT

Itemized list of deliverables

1. **Project Documentation:** One completed project proposal, literature review and final project report.
2. **System Requirements Specification:** Functional and non-functional requirements.
3. **System Design:** User interface design, system architecture, and database design.
4. **Functional Web Application Prototype:** It is an application that is locally executable and has separate interfaces for patients, doctors, and the document review team.
5. **Document Review Module:** Document submission and review, decision-making, and requests for correction.
6. **Patient Health Record Module:** Approved health records organized chronologically, and patient-controlled doctor access.
7. **Consultation Summary Feature:** A summary created from a selected set of patient records for use in preparation for a consultation.
8. **Testing and Evaluation Report:** Test cases, results, and usability evaluation findings using fictitious data.
9. **Final Project Demonstration:** A demonstration of the application running locally and the key functionality of the application, including screenshots and technical documentation.

CHAPTER 02

LITERATURE REVIEW

Personal Health Records

Personal Health Record (PHR) systems allow individuals to store, manage and access their health records electronically. Roehrs et al. (2017) point to the role of PHR systems in augmenting patients' participation and access to health information. Properly arranging and keeping medical documents and records in chronological order allow users to easily access their medical history and provide pertinent information when consulting with others. But usability, data security, and effective information management are key factors in designing successful PHR systems.

Security and Information Sharing

Access and controlled sharing of information are vital in a digital healthcare system. Access control and privilege management in EHRs is highlighted as important in the work of Jayabalan and O'Daniel (2016). Permission mechanisms are in place to ensure the safekeeping of sensitive information but also for authorized persons to be able to retrieve relevant records. Hence, one of the critical issues in developing a PHR system to enhance privacy and secure information sharing is patient control of doctor access.

Analysis of Existing Systems

There are good reference points for understanding digital health information management from existing digital healthcare platforms, such as HealthSync, NexusCare and Medica. HealtSync offers help with documents to upload and sharing health records, digital healthcare services are represented by NexusCare and Medica. The significance of accessible healthcare solutions in the digital domain is highlighted by these platforms. The information available in the public domain, however, does not substantiate whether these systems offer the same combination of

document submission, administrative review, chronological record management and doctor access to patients as are suggested in this project.

Research Gap and Proposed Contribution

The literature reviewed emphasizes the need for personal health records to be accessible, secure information sharing, and having proper access control. The envisioned Personal Health Record Management and Sharing System combines document submission, administrative review, chronological document management, and doctor access to the patient's records, all in the same prototype, based on these principles. The system has a goal of offering an structured workflow for handling and sharing personal health data. The role of the review team is to review the administrative documentation, not to be a clinical verification. The prototype will be tested for its functionality and usability with sample data.

Summary

Studies in the literature suggest that good PHR systems should enable the organization of health information, provide secure access and encourage patient involvement. The proposed project is conceived as an integrated prototype that will enhance the support of PHR management and controlled information sharing in the field of personal health records.

CHAPTER 03

RESEARCH AND DEVELOPMENT METHODOLOGIES

Research Methodologies

The project will take an applied research path to uncover the needs of the users, and create a feasible solution for managing personal health records. Data will be collected by analyzing literature and current digital health platforms. The results will help determine the system requirements, such as who is going to submit documents, how documents will be reviewed by administration, how patients will access their health records, and how to organize the health records. The final prototype will undergo functional testing and – if possible – will receive usability testing input based on fictional data.

Development Methodology

The application will be developed in an iterative way in manageable steps. This allows the separate features to be developed, tested and refined prior to being put together as the entire system.

The development process will include:

1. **Requirements Analysis:** Identify the patients, doctors, and document review team requirements.
2. **System Design:** Develop interface designs, system architecture, and database structure.
3. **Implementation:** Create the user interfaces, document flow for the review process, health record management, and access to documents.
4. **Testing:** Test core functions, role-based access, and handling of records with fictional data.
5. **Assessment and Improvement:** Examine the test reports, correct the found problems, and enhance usability if possible.

The application will be developed and demonstrated in a local environment. Online hosting, public deployment, and real healthcare system integration are outside the project scope.

TECHNOLOGIES & RESOURCES

Technology

The proposed system will use a suitable frontend and backend technology to develop the web application. Python with Flask, HTML, JavaScript and Bootstrap are potential options, while MySQL or another suitable database may be selected for data storage. System architecture and design diagrams will be prepared using an appropriate diagramming tool. The technology stack may be adjusted during development based on technical requirements and compatibility.

Hardware / Software Resources

Hardware Resources

- Laptop or desktop computer
- Sufficient RAM and storage for development and testing
- Internet connection for research and downloading development tools

Software Resources

- Windows operating system
- Python and Flask (if needed)
- HTML, JavaScript, and Bootstrap
- MySQL or another suitable database
- Visual Studio Code or another code editor
- Web browser
- Draw.io for system design diagrams
- GitHub for version control, if required
- Microsoft Word for project documentation

This version adopts relevant tools from your example but excludes cloud storage, virtual-machine recovery environments, backup recovery testing and RTO/RPO measurements because they are not part of your proposed health-record application.

PROJECT PLAN

Gantt chart / Timeline

Gantt Chart – MIS 4500 Technology Capstone Project

Fall 2026 | Semester start: Mon 21 Sep 2026

Activity	September 2026		October 2026				November 2026				December 2026	
	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
	21 Sep – 27 Sep	28 Sep – 04 Oct	05 Oct – 11 Oct	12 Oct – 18 Oct	19 Oct – 25 Oct	26 Oct – 01 Nov	02 Nov – 08 Nov	09 Nov – 15 Nov	16 Nov – 22 Nov	23 Nov – 29 Nov	30 Nov – 06 Dec	07 Dec – 13 Dec
Class days (Tuesday and Thursday)	Tue 22 Sep Thu 24 Sep	Tue 29 Sep Thu 01 Oct	Tue 06 Oct Thu 08 Oct	Tue 13 Oct Thu 15 Oct	Tue 20 Oct Thu 22 Oct	Tue 27 Oct Thu 29 Oct	Tue 03 Nov Thu 05 Nov	Tue 10 Nov Thu 12 Nov	Tue 17 Nov Thu 19 Nov	Tue 24 Nov Thu 26 Nov	Tue 01 Dec Thu 03 Dec	Tue 08 Dec Thu 10 Dec
Assessed milestones		1	2	3			4				5	6
Project orientation and planning												
Requirements analysis												
Literature review and system research												
System design and database planning												
User interface development												
Core functionality development												
System integration												
Functional testing												
Usability and user evaluation												
Refinements and bug fixing												
Final report and documentation												
Final demonstration and preparation												

Planning · Development · Testing and evaluation · Documentation and delivery · Assessed milestone

1 **Week 2:** Title Selection – 10%
2 **Week 3:** Proposal Presentation – 10%
3 **Week 4:** Proposal Submission – 10%
4 **Week 7:** Project Demo – 20%
5 **Week 11:** Oral Viva and Demo – 20%
6 **Week 12:** Project Report Submission – 30%

Course weeks run Monday to Sunday; Week 1 begins on the semester start date (Mon 21 Sep 2026), with the first class on Tue 22 Sep. Weeks are grouped under the month of their Tuesday class. Milestones are shown within their course week because the exact class day is not specified. Weights total 100%.

CONCLUSION

This project aims to create a web-based Personal Health Record Management and Sharing System which would enable patients to manage their health records and share their authorized health records with physicians. It will streamline medical document submission, administrative review, health record management, and create consultation-ready summaries in a separate interface for patients, doctors, and the document review team.

Development will follow the principles of a functional prototype and will be tested locally with fictitious data. It will be tested for effectiveness using functional testing and usability assessment. The outcome shall be a real-life example of an integrated workflow that can facilitate coordinated health information management and controlled access to patient records in the scope of the project.

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